A file is a named collection of related information that is recorded on secondary storage such as magnetic disks, tapes & optical disks.
A file is an abstract data type.

* Unit VI: File Management [10M]

· File attributes

(i) name - The symbolic file name is the only information kept in human readable form.

(ii) identifier - file system gives a unique tag or number that identifies file within file system & which is used to refer files internally.

(iii) type - This information is needed for those systems that support different types.

(iv) Location - This information is a pointer to device & to location of file on that device.

(v) Size - The current size of file and possibly maximum allowed size are included in this attribute.

(vi) Protection - Access control information determines that who can do reading, writing, executing & so on.

(vii) Time, Date & User Identification - This information may be kept for creation, last modification & last use. These data can be useful for protection, secu...

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• File operations

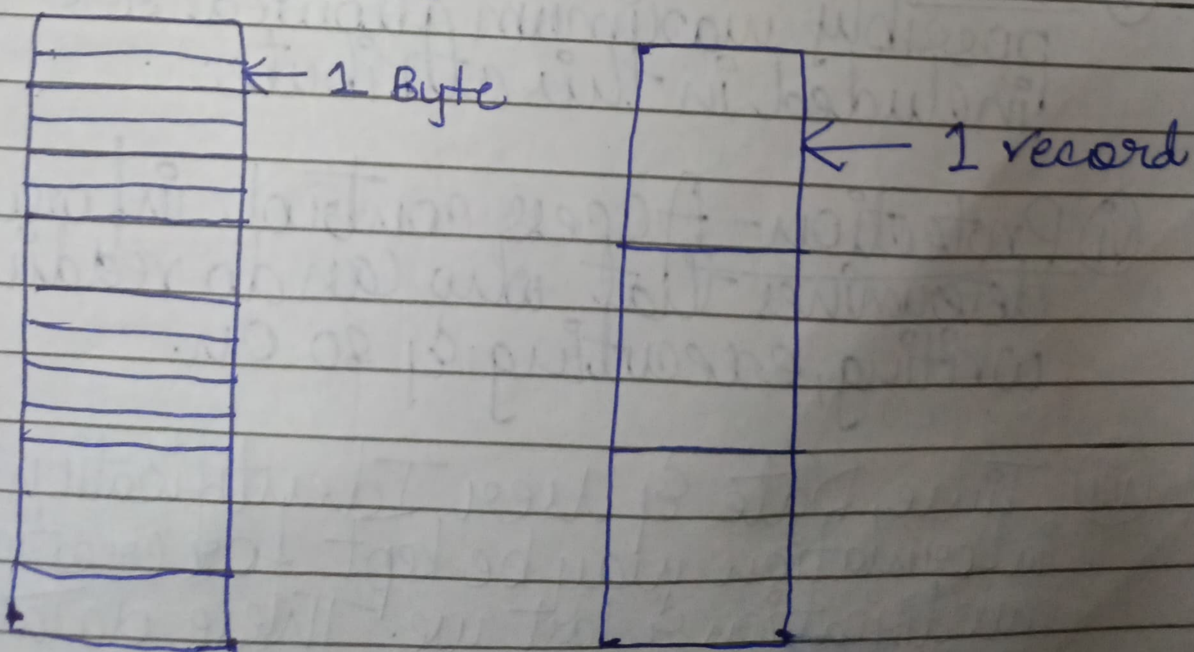
(i) **Creating a file** - Two steps are necessary to create a file. First space in file system must be found, second an entry for new file must be made in directory.

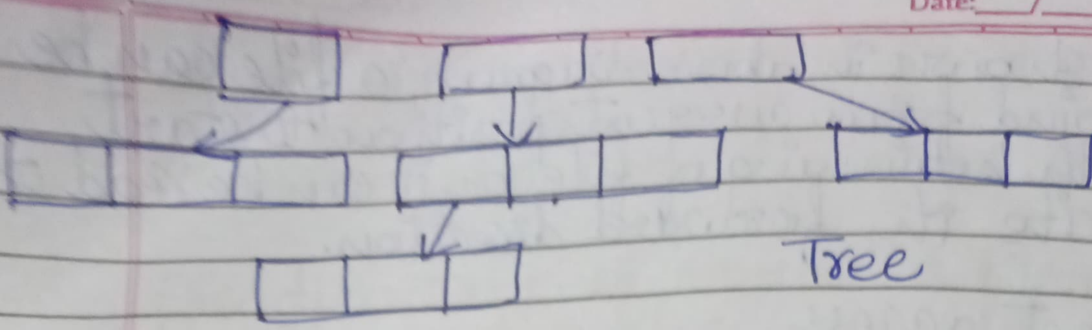
(ii) **Writing a file** - To write a file, we make system call, specifying both name of file & information to be written to file.

(iii) **Reading a file** - To read from a file, we use system call that specifies name of file & where next block of file should be put.

(iv) **Deleting a file** - To delete a file, we search directory for named file. Having found associated directory entry, we release all file space & erase directory entry.

• File structure





Tree

- Access methods

(i) Sequential access:

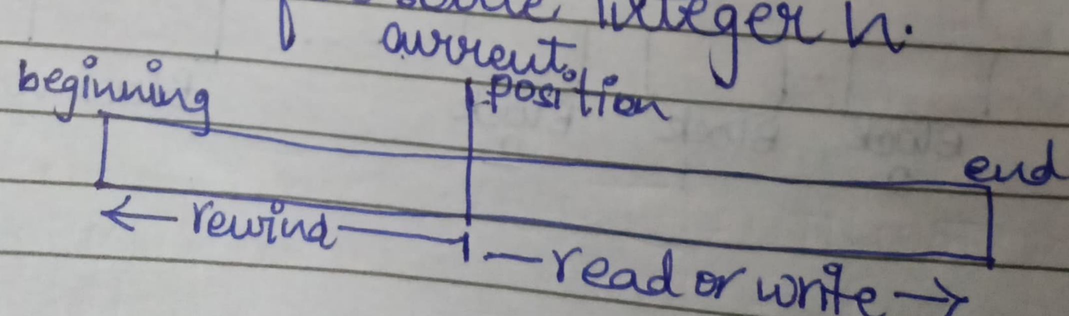
(i) Information from file is processed in order i.e. one record after another and is commonly used access mode.

(ii) For example, editors and compilers access files in sequence.

(iii) A read operation read information from file in a sequence i.e. read next reads next portion of file and automatically advances a file pointer.

(iv) A write operation writes information into file in a sequence i.e. write next appends to the end of file and advances to the end of newly written material. Such file can be reset to the beginning.

(v) In some operating systems, a program may be able to skip forward or backward 'n' records for some integer n.



As shown in above diagram, a file can be rewind from current position to start with beginning of file or it can be read or write in forward direction.

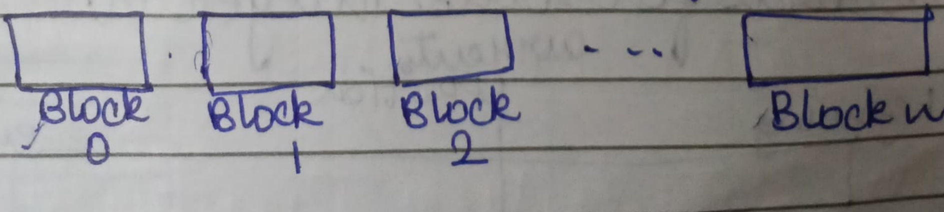
② Direct access

(i) It is also called as relative access. A file is made up of fixed length logical records that allow programs to read and write records rapidly in no particular order.

(ii) This method is based on disk model of a file which allows random access to any file block.

(iii) For direct access a file is viewed as numbered sequence of blocks or records. So we can directly read block 14, then block 53 and so on. This method is used for immediate access to large amount of information.

(iv) Database can be accessed with direct access method. For example, when a query concerning a particular subject arrives, we ~~ys~~ compute which block contains the answer and then read that block directly to provide the desired information.



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Advantages of sequential file access:
(i) Easy to access the next record
(ii) Data organization is very simple

Disadvantages of sequential file access:
(i) Provides poor performance
(ii) More time consuming since reading, writing & searching always start from beginning of file.

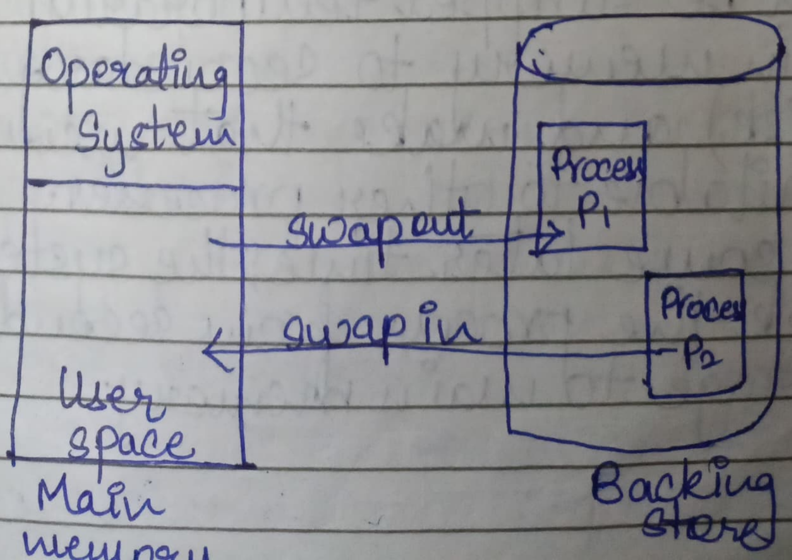
Advantages of direct access:
(i) Any record can be accessed randomly
(ii) Gives fastest retrieval of records

Disadvantages of direct access:
(i) Method is complex
(ii) More expensive

- Swapping
(i) It is a mechanism in which a process can be swapped temporarily out of main memory to secondary storage (disk) and make that memory available to other processes.
(ii) At some later time, the system swaps back the process from secondary storage to main memory.

(iii) For example, assume a multiprogramming environment with round-robin CPU scheduling algorithm. When a quantum expires, the memory manager will start to swap out the process that just finished and to swap another process into memory space that has been freed. In meantime, CPU scheduler will allocate a time slice to some other process in memory. When each process finishes its quantum, it will be swapped with another process.

(iv) A variant of this swapping policy is used for priority-based scheduling algorithms. If a higher-priority process arrives and wants service, the memory manager can swap out lower-priority process and then load & execute higher-priority process. When higher-priority process finishes, lower-priority can be swapped back in and continued. This is sometimes called roll out, roll in



- Swap-out is a technique for moving process from RAM to hard disc.
- Swap-in is a method of transferring a program from hard disc to main memory or RAM

• File Allocation Methods

(i) From user's point of view file is an abstract data type. The direct-access nature of disks allows us flexibility in implementation of files.
(ii) In almost every case, many files are stored on same disk. The main problem is how to allocate space to these files so that disk space is utilized effectively and files can be accessed quickly.
(iii) Three major methods of allocating disk space are in wide use: contiguous, linked and indexed.

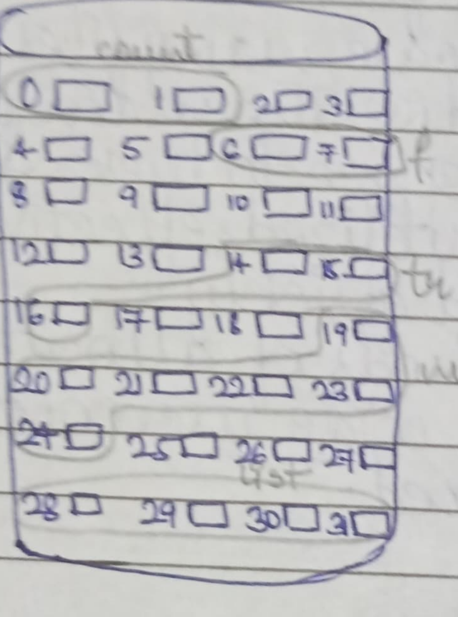
(1) Contiguous.

a) This allocation requires that each file occupy a set of contiguous blocks on the disk. Disk addresses define a linear ordering on disk.
b) Contiguous allocation of file is defined by disk address and length (in block units) of first block. If file is 'n' blocks long and starts at location 'b', then it

occupies blocks $b, b+1, b+2, \ldots, b+n-1$.

c) The Accessing file that has been allocated contiguously is easy for sequential access, file system remembers the disk address of last block referenced and when necessary reads the block.

d) For direct access to block i of a file that starts at block b, we can immediately access block $b+i$. Thus, both sequential and direct access can be supported by contiguous allocation.



Directory

File	Start	Length
count	0	2
tr	14	3
mail	19	6
list	28	4
f	6	2

e) Advantages:
- Supports both sequential & direct access methods.
- Provides good performance

f) Disadvantages:
- Suffers from external fragmentation
- Compaction may be required and it can be very expensive.

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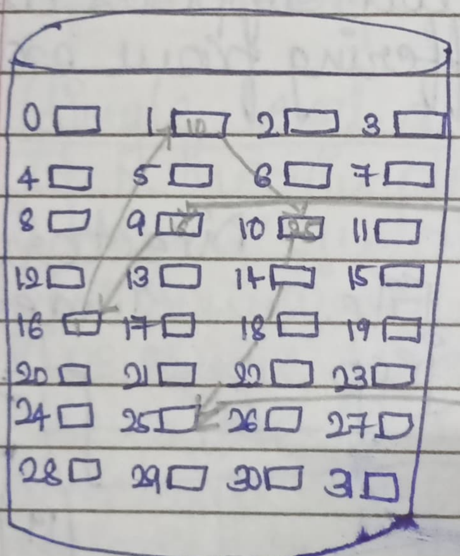
② Linked

a) With linked allocation, each file is a linked list of disk blocks; the disk blocks may be scattered anywhere on the disk.

b) The directory contains a pointer to first and last blocks of file. Each block contains a pointer to next block. These pointers are not made available to user.

c) To create a new file, simply create a new entry in directory.



Directory

File	Start	End
jeep	9	25

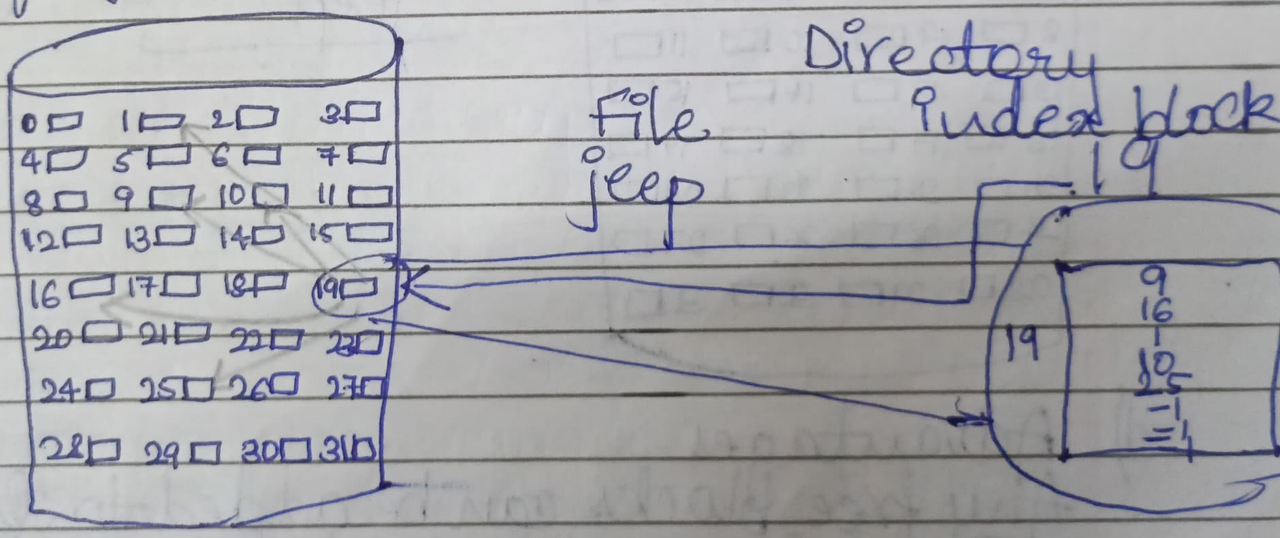
d) Advantages
- Any free blocks can be added to chain
- There is no external fragmentation
- Best suited for sequential files that are to be processed sequentially.

e) Disadvantages
- Does not support direct access

- Requires more space to store pointers.

③ Indexed

a) Indexed allocation ~~solves~~ brings all of the pointers together into one location the Index block.
b) Each file has its own index block which is an array of disk block addresses. The i^{th} entry in index block points to i^{th} block of file. The directory contains address of index block.
c) Indexed allocation supports direct access, without suffering from external fragmentation.



d) Advantages
- Support both sequential & direct access
- Does not suffer from external fragmentation

e) Disadvantages
- Keeping index in memory requires space.
- Overhead of index blocks is not feasible.

- Directory structures

A directory is a container that is used to contain folders & files. It organizes files and folders in a hierarchical manner. Following are logical structures of directory:

① Single-level directory
(i) It is the simplest form of directory system is having one directory containing all files. Sometimes, it is called root directory.
(ii) In single level directory structure, the entire files are contained in same directory. So unique name must be assigned to each file of directory.
(iii) It was implemented in older versions of single user systems. The world's first supercomputer, the CDC 6600, had only single directory for all files, even though it was used by many users at once.
(iv) Diagram:

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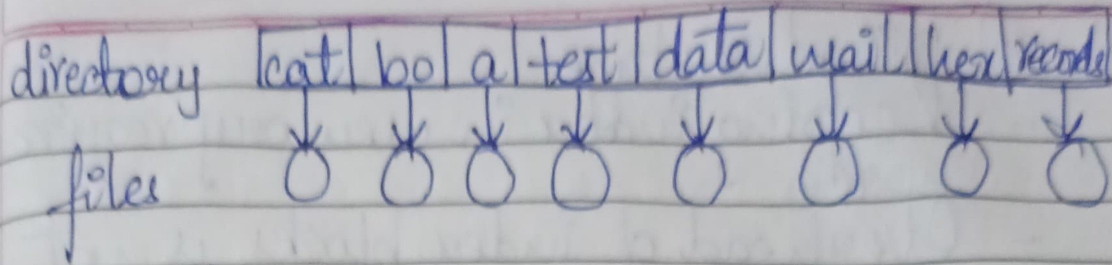

(v) Advantages:

a) Implementation is very easy

b) Operations like file creation, searching, deletion, updating are very easy in such a directory structure.

(vi) Disadvantages:

a) There maybe chance of name collision because two files can have same name.

b) Searching will take time if directory is large.

2) Two-level directory

(i) The structure of two level directory structure is divided into two levels of directories namely, a master directory & user directories. User directories are sub-directories of master directory.

(ii) In two level structures, each user has its own user file directory (UFD). The UFD lists only files of single user.

(iii) System contains a master file directory (MFD) which is indexed by username or account number. Each entry in MFD points to UFD for that user.

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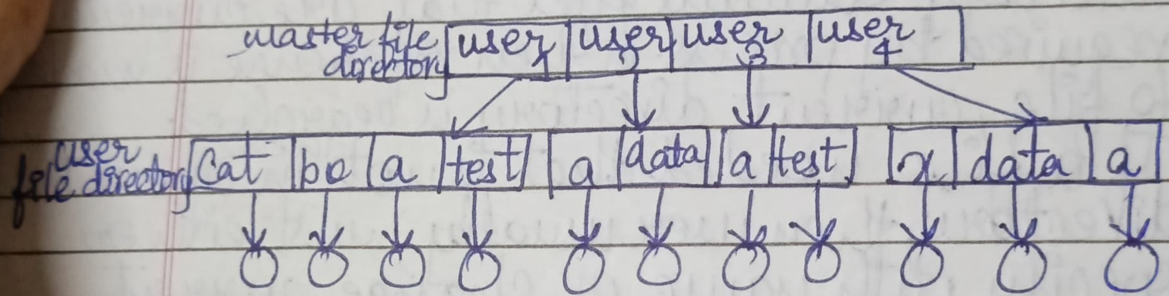
(iv) When a user refers to particular file, only his own UFD is searched. Different users can have files with same name, as long as all file names within each UFD are unique.

(v) When we create a file for user, operating system searches only that user's UFD to find whether same name file already present in directory. For deleting a file again operating system checks file name in UFD only.



(vi) Advantages:

a) There can be more than two files with same name and would be helpful if there are multiple users.

b) Searching of files becomes easy

(vii) Disadvantages:

a) Users don't have ability to create subdirectories

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(3) Tree / Hierarchical directory structure

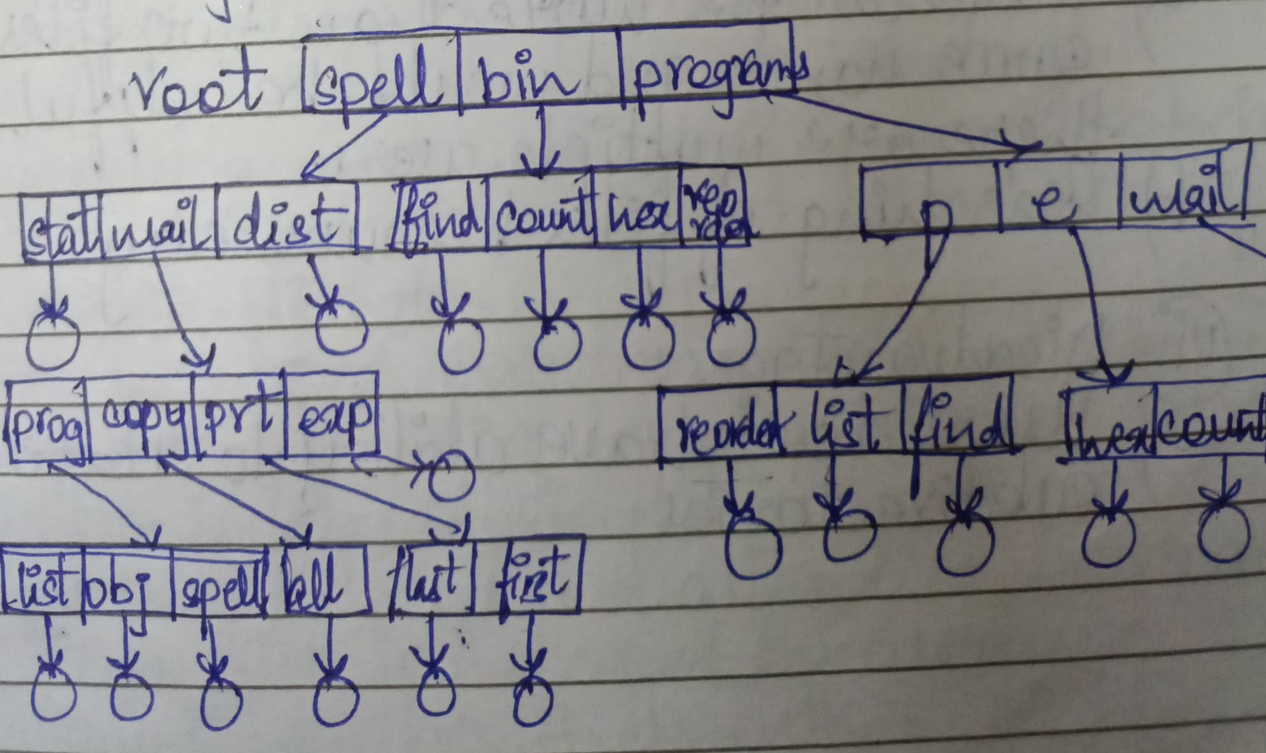
(i) In this directory structure user can create their own sub-directories and organize their files.

(ii) The tree has root directory and every file has unique path name. A directory contains set of files or subdirectories.

(iii) All directories have same internal format. One bit in each directory entry defines entry as file (0) or as subdirectory (1)

(iv) Each process has current directory. Current directory contains files that are currently required by process. When reference is made to file, current directory is searched.

(v) If a file needed that is not there in current directory, then user usually must either specify path name or change current directory.



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Advantages
Allows subdirectories inside directory.
More scalable
Disadvantages.
Users cannot modify root directory data.